

Guidelines for Write-Ups on the “Littlefield Labs Simulation”

Each team has to turn in a three-page write-up (and append any number of graphs and tables necessary). In this write-up you have to present a summary of all actions you took during the simulation (all that information can be downloaded by accessing your account), provide explanations of the reasons for taking those actions, and in retrospect whether you think you did the right thing (in other words, if you were playing the simulation for a second time what would you have done differently).

The main issues to be addressed in the write-up are setting of capacity levels at each station, simple logic of inventory control used during the game, both observed and desirable target levels of utilization of the different resources, identification of bottleneck resource(s), and choice of priority scheduling rule. The actions you take during the simulation are to buy new machines, sell existing machines, set reorder point and reorder quantity, choose contract type offered to customers, adjust priority-scheduling rule, and change debt. Please review the history of your simulation and try to explain the reasons for each action you took. The provided explanations should be addressing in adequate detail the above main decision issues in the simulation.

We would prefer to see some quantitative analysis in support and justification of your actions. However, much more important for us is your intuitive understanding of the “physics” of the system and the logic of managing it. You should be aware that there are options in the simulation program to plot graphs of utilization, queue sizes, lead times and revenues. You can also easily download such data for spreadsheet analysis during or after the simulation. To present quantitative justification of your decisions, feel free to attach tables or graphs based on these spreadsheets to your write-up. Of course, feel free to use the “decision support” spreadsheet we provided you during the simulation. Pages of tables and graphs are not counted in the three-page limit.

In all likelihood you will make your early simulation decisions using simple intuition, with some of it, hopefully, derived from course concepts. We repeat for one more time, we highly value your commenting on the guiding intuition of your decisions in your write-up. After the completion of the simulation, you will have more time to think about it, and that might allow you to do more quantitative analysis. By performing appropriate calculations after the simulation, you can always in retrospect assess if you made the appropriate decisions given the available information, and if you were to rerun the simulation and repeat the decisions what changes you would have made.

Your team's grade is partially based on your team's overall rank based on revenue performance. However, the largest weight is given to the presented analysis, understanding of decision tradeoffs, and appropriate decision logic in your write-up. Once again, high quality write-ups will be characterized by some quantitative analysis, very clear intuitive explanations of the observed system, and managerial insights on how to manage it applying appropriately concepts from our operations management course.

We hope that you fully enjoy the simulation and also have a great learning experience!!